QUALCOMM pdQ Smartphone
Imagine the Possibilities
Session 524



Eydie Galper

Extending the Palm Computing® Platform

The Palm Computing® Platform
Intuitive, Functional, Popular



CDMA Voice and Data
Fastest Data Rate, Lowest Cost, Best Security



Palm + CDMA Wireless Phone

Integration & New Applications

Over-the-Air Data

Developer Interfaces



The Premier Integrated Wireless Data Device



Integration & New Applications

- Enhanced Address Book
- Speed Dial
- Call History
- pdQbrowser
- Eudora
- pdQalert
- Dialer



pdQ Phone Specifications

- Two Models Available
  - 1900 Digital
  - 800 Dual Mode
- Battery Life - 2.5 Hours Talk Time, 50 Hours Standby
- Internal Backup Battery
- Contact Your Wireless Carrier for Purchasing Information



pdQ Smartphone
An Internet & Telephony Platform



Marc Weiss

Overview

- Developer Basics
- Telephone Capabilities
- Getting Connected
- Built-in Internet Applications
- URL Handling Features
- Q & A



Developer Basics

- HTML Web Development
- Standard Palm OS Development
- pdQ Telephone and Internet Capabilities



Palm OS™ Compatibility

- Based on Palm OS 3.02
- Equivalent to Palm III
- Applications Using Standard Palm OS Calls Will Work
- Non-Palm API Calls Will Most Likely Fail (i.e. Direct Screen Access)



Telephone Capabilities

- Make a Call or Answer a Call
  - Dial From the Address Book
- Phone State Notification
  - Call History for Incoming and Outgoing Calls
- Flash LEDs, Turn Backlight On and Off



pdQ Development Zone

- SDK Available Online
- www.qualcomm.com/phones/products/pdq_phone
- pdQ Mailing Lists
- Developer Email Questions
  pdQdeveloper@qualcomm.com



QUALCOMM

pdQ Development Lab
Thursday October 21, 1999



Session 188
2:15PM - 3:15PM

Getting Connected With pdQ Data



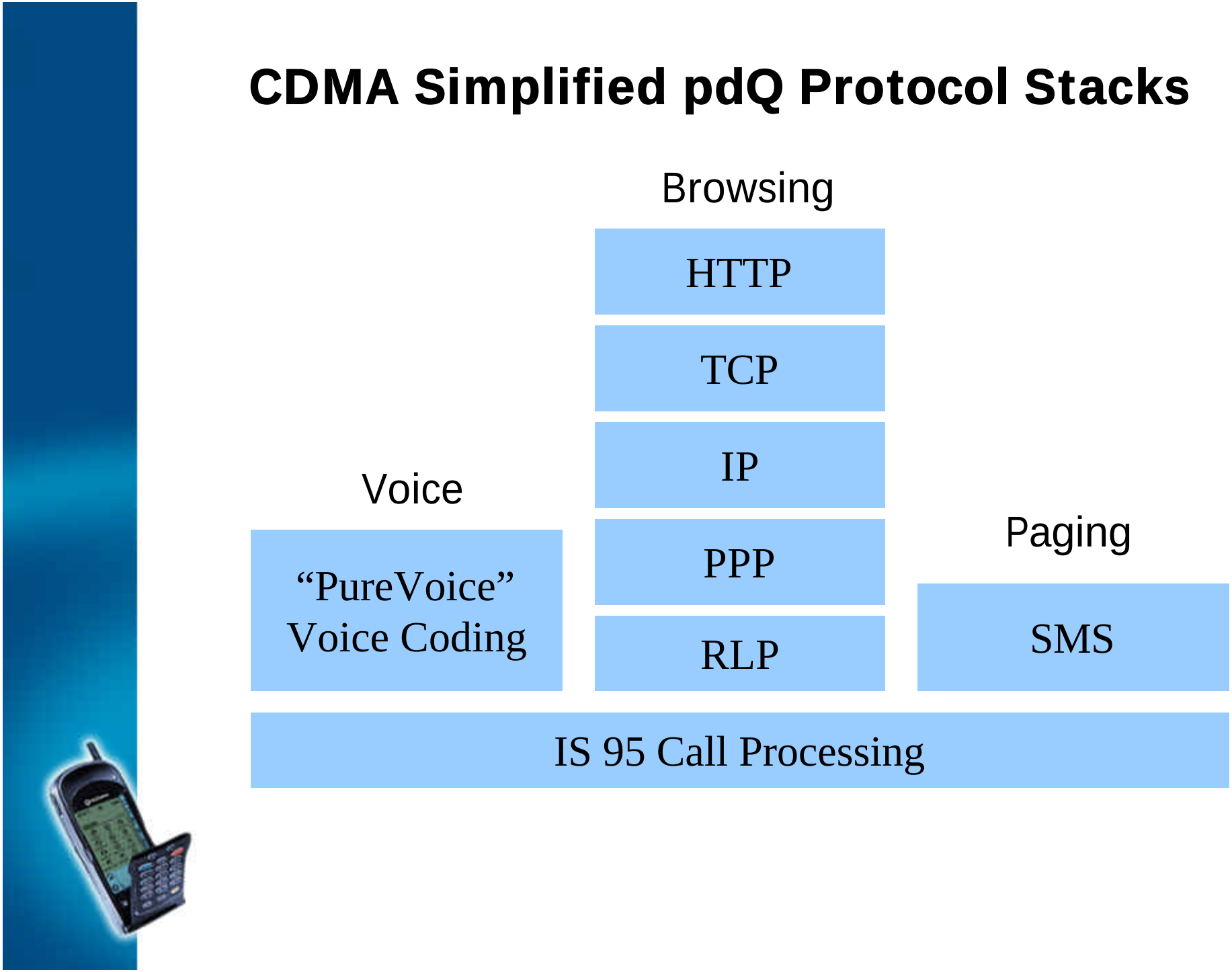
Laurence Lundblade

CDMA Wireless Networks

- Most Recent Technology Generation for Wireless Voice and Data Communication
- Replaces AMPS (Analog) and TDMA
- CDMA Carriers in US Are Sprint, Airtouch, Bell Atlantic, PrimeCo, and Others
- CDMA Provides Data Services, Increased Capacity, Improved Voice Quality, Better Hand-offs, Basic Security, Lower Transmit Power and Better Battery Life



CDMA Simplified pdQ Protocol Stacks



pdQ Data Performance

- Throughput Is 14.4kbps
- Higher Rates on the Way
- Can Transfer 50K Bytes in Under a Minute
- TCP Connect to Host Is 2-10 Seconds (Depends on How Busy Host Is)



pdQ Data Connection

QNC	Async / Circuit Switched
Wireless carrier is your ISP	Wireless carrier dials out for you to an ISP
Connect time: 5-20s	Connect time: 20-40s
Wireless carriers assigns IP address	ISP/Corporate dial-in assigns IP address
Outside corporate firewall	Inside corporate firewall

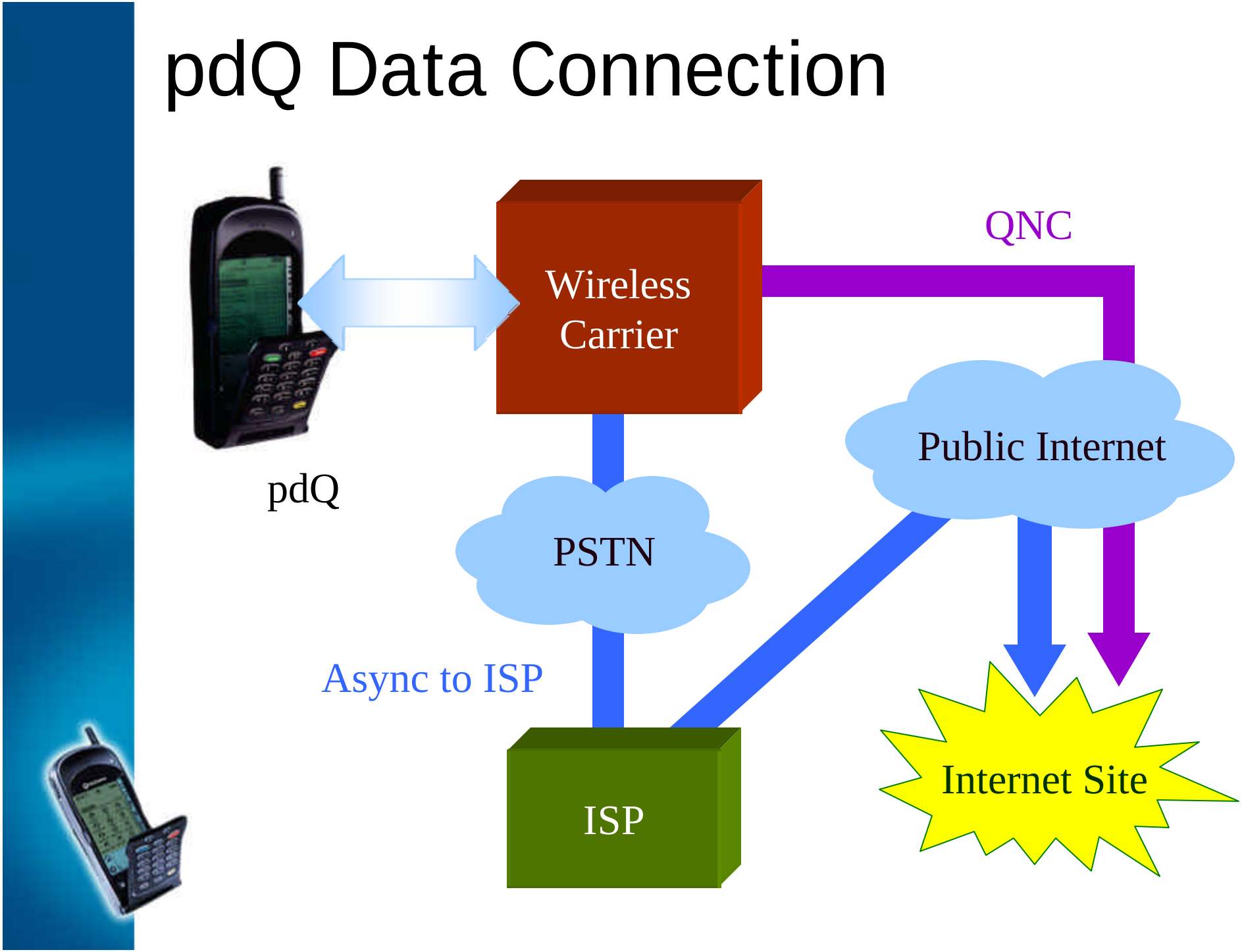


pdQ Data - Connection

- Can Connect to Any Host on Public Internet
- Can Use **ANY** TCP or UDP Based Protocol (e.g., HTTP, ICQ, SMTP, Telnet, DNS)
- CDMA Coding Provides Basic Over the Air Security



pdQ Data Connection



pdQ Corporate Access



pdQ

Wireless Carrier

Internet Site

PSTN

Public Internet

Firewall

Corporate Data

Corporate Dialup

Async through Corporate Dialup



pdQ Data - Costs

- Connection Charges Are Set by Your Wireless Carrier
- Data Typically Billed As a Voice Call 0.20/min.
- Transferring 50K Bytes Costs Less Than $0.20
- Less Expensive Than Other Wireless Data Networks Because Infrastructure Largely Paid for by Voice Traffic. (Antennas already up; Systems in place)



Using Data Services

- Use Standard TCP/IP calls Same As for Palm Modem
- Straight Socket Programming
- Opening Netlib Starts a Data Call (Like a Land Line Palm Modem Connection)
- Network Preferences Come From Carrier
- If You Don't Have a pdQ, Write Your Application Using a Palm With a Modem.



Connectivity Summary

- CDMA Data Is Available Worldwide
- CDMA Data Available From Multiple Carriers
- Good Throughput and Low Cost
- Develop Using Palm's Standard Netlib
- Secure



pdQ Built-In Internet Applications



Built-In Internet Applications

- Email — Eudora
- Web Access — pdQbrowser



Email — Eudora

- Standard POP3/SMTP
- Handles All Common Message Formats Including MIME and HTML
- Can Tap on URLs in Email Messages
- Handles mailto: URLs
  - Can Be Launched by Other Apps
  - Fields Supported Include: To, CC, Subject, Body



Web Access — pdQbrowser

- Standard HTTP
- No Proxy Needed
- Authentication
- Cookies



Web Access — pdQbrowser

- Supports HTML 3.2
- Text-Only Browsing — No Graphics
- Forms Input Elements Displayed In-line
  - TEXT, PASSWORD, CHECKBOX, RADIO, SUBMIT, RESET, HIDDEN, (IMAGE)
  - Forms Also Support TEXTAREA & SELECT
  - HTML Frames Are Minimally Supported
- Specifications in pdQ SDK Document



Internet End-to-End

- Standard Internet Applications Ship in ROM
- No Proxies Are Needed
- pdQ HTML tag available
- Develop for Internet Standard Protocols and Formats
- Platform Is Capable Enough to Do Standard Internet Applications



pdQ URL Handling



pdQ URL Support

- Can Send http: URLs to pdQbrowser
- Can Send tel: URLs to Dialer
- Can Send mailto: URLs to Eudora
- Can “Speed Dial” URLs!
- Use the pdQ URL Registry to Launch a URL Instead of Direct Application Switch.



pdQ URL Registry

- Maps a URL Scheme (e.g. “http:”, “mailto:”) to a Palm Application
- Used by pdQbrowser, Eudora, Address Book and Other Applications
- Any Application Can Register for Any URL Scheme
- Developer can create custom schemes



Custom URL Registry Example

- You Are a Stock Web Site and You Want to Offer Advanced Quote Information to pdQ Users. (stock graphing feature)
- The Developer…
  - Creates a Palm Application That Displays a Graph of Stock Prices (.prc)
  - Registers the Palm Application for the “stock-graph:” URL Scheme
  - Creates a pdQ Area on the Web Site With a “Show Graph” Button Linked to the “stock-graph:” URL



Custom URL Registry Example

- "Show Graph" Button Links to **<A HREF="stock-graph:qcom, today,170,171.5,175,181, 182,198,202…">** URL
- The User…
  - Visits Web Site Using pdQbrowser
  - Pushes "Show Graph" Button in pdQbrowser and Gets a Graph of the Stock Activity



Custom URL Registry Demo



Custom URL Notes

- This Example Puts All Data in URL for Simplicity
- URL Could Use Any Protocol to Send or Receive Larger Amounts of Data
- Other Examples:
  - Fancier Graph (e.g., 3-D Rotation/modeling)
  - Custom Input Control (e.g., Input a Date)
- Custom URLs Work with Eudora and Other Apps That Launch Through the Registry



pdQ Is a Phone
pdQ Is a Palm Computing® Platform Organizer

- A Phone That You Can Programmatically Control
- Good, Reasonably Priced Internet Connectivity
- Internet Standard Through and Through
- Secure



Q & A

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